One Stack, Many Rankings: Evaluating Quantized Reasoning Beyond Accuracy

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Abstract

Post-training quantization is a common serving default for reasoning language models, but a single pass@1 or token count is not a complete evaluation. We evaluate public R1-Distill checkpoints under one pinned serving stack—vLLM 0.7.0 [11] eager execution on an NVIDIA A100-80GB, with FP8 checkpoints executed as Marlin W8A16 [5] rather than native W8A8—and vary only the public weight checkpoint among BF16, FP8, AWQ-4, and GPTQ-4 for DeepSeek-R1-Distill-Qwen-7B and DeepSeek-R1-Distill-Llama-8B. The evaluation comprises 88 checkpoint×benchmark×seed runs and 56,408 completions on MATH-500 [8, 14] (5 seeds), GSM8K [2] (3 seeds), and GPQA-Diamond [19] (3 seeds).

Under this pin, estimands disagree. On MATH-500, FP8–BF16 pass@1 differences are +0.40 and +0.28 percentage points (pp); problem-clustered 95% intervals include zero, and a ±1 pp equivalence test is not passed. The tested Qwen AWQ-4 checkpoint loses 5.56 pp on GPQA-Diamond (Holm-significant within the primary six GPQA contrasts, not under a joint 18-contrast Holm sensitivity). Mean MATH-500 length rises 6.3–6.9% for the tested Qwen 4-bit checkpoints, including detectable lengthening among jointly correct Qwen AWQ-4/GPTQ-4 pairs; much larger conditional differences occur in mismatch cases, where failure traces are themselves long. Strict five-sample unique-mode agreement yields low observed selective risk, trading off coverage and requiring five generations. The historical token proxy, sequential Condition A, and batched Condition B produce different rankings: the proxy ranks Qwen FP8 first among Qwen cells, Condition A ranks Qwen AWQ-4 first, and Condition B ranks Qwen GPTQ-4 first. Effects and rankings are checkpoint-, task-, workload-, estimand-, and serving-condition-specific under this pinned stack.

Keywords: Reasoning language models, public quantization checkpoints, pinned serving stack, estimand disagreement, Cost-of-Pass.

1 Introduction

Reasoning models such as DeepSeek-R1 [7] emit long traces before a final answer. Post-training quantization (FP8, AWQ-4 [15], GPTQ-4 [4]) is therefore a common serving choice. Isolated pass@1 numbers hide whether compression also changes trace length [13, 17, 18], termination, or the cost of a correct answer [3, 10]. Serving-stack details—engine version, decoding flags, CUDA graphs—can materially move reasoning scores [9]. This paper *pins* one stack rather than shifting it: we do not run a factorial vLLM 0.7.0 vs. 0.8.5 experiment.

We ask four questions under that pin:

- **RQ1.** Does quantization change *pass@1* relative to matched BF16, with problem-clustered uncertainty?

- **RQ2.** How do completion length and correctness-conditioned length differ across the evaluated checkpoints, and what do identical-word loops and near-cap completions reveal about the long-tail behavior?
- **RQ3.** What can be said about multi-sample reliability without gold labels at serve time?
- **RQ4.** Do checkpoint rankings agree across the historical token proxy, sequential Condition A, and batched Condition B aggregate serving-cost proxies?

Contributions. (C1) A pinned protocol with problem-clustered pass@1, ± 1 pp TOST, and an explicit distinction from maj@5 consensus. (C2) Checkpoint-specific accuracy: the tested Llama AWQ-4 checkpoint shows the largest MATH-500 and GSM8K pass@1 drops among the evaluated Llama cells; the tested Qwen AWQ-4 checkpoint shows the largest GPQA-Diamond drop among the evaluated Qwen cells; FP8–BF16 MATH intervals include zero but fail TOST. (C3) Length estimands on the full MATH-500 grid: ratio of means, 2×2 correctness strata with clustered CIs, a clustered mismatch-excess contrast D, a BF16 failure-length control, and a BF16-correct conditional Δ, following Lian et al. (C4) Post-hoc gold-free unique-mode abstention on five MATH-500 seeds, with Wilson intervals on selective risk. (C5) Cost-estimator disagreement: rankings from a historical 65 tok/s token proxy, sequential Condition A, and batched Condition B aggregate hybrid Cost-of-Pass proxies do not agree. The released grid and analysis artifacts support reproducibility of these contributions.

2 Related Work

GPTQ [4] and AWQ [15] were originally validated mainly on short-form benchmarks. SmoothQuant [21] addresses activation quantization, which we do not run. Liu et al. (QRM) [17] evaluate DeepSeek-R1 distillations across weight, activation, and KV quantization. QRM reports that *minor-drop* W8/W4A16 settings do not systematically lengthen outputs, whereas more aggressive low-bit settings can. Numerically, QRM reports Qwen-7B MATH BF16 94.60 / AWQ 91.00 and GPQA AWQ 50.00; this pinned grid is 93.12 / 44.78 for the tested Qwen AWQ-4 checkpoint. We therefore interpret the cross-study discrepancy as evidence of checkpoint- and stack-sensitive measurement rather than as a superiority claim. Lian et al. [13] isolate a BF16-correct length estimand: low-bit PTQ can inflate chain-of-thought even when the uncompressed run is right. Lotfi et al. [18] attribute much of the extra length to overthinking—abandoned intermediate answers rather than missing competence. Li et al. [12] localize 4-bit math failures to early process errors. Alimaskina et al. [1] study loops and delayed commitment under extreme low-bit inference. Kurtic et al. [10] measure accuracy–throughput trade-offs for Llama-3.1 quantization in vLLM across deployment regimes; we ask the narrower question of whether *rankings* among eight public R1-distill checkpoints agree across cost estimators on one A100/vLLM 0.7.0 pin.

Hochlehnert et al. [9] show that seeds, prompts, decoding, and software move reasoning scores; we freeze those factors. Zhong et al. [22] study uncertainty calibration of quantized LLMs using model probabilities—a quantity we do not log. Self-consistency [20] and G-Pass@k [16] are gold-using repeated-sampling neighbors; our modal rule is a gold-free unique-mode abstention on five MATH seeds, not a new estimator. Selective classification [6] supplies the risk–coverage language. Erol et al. [3] define Cost-of-Pass as expected monetary cost per correct solution. We report a historical shared-65 tok/s token-implied aggregate cost proxy and an aggregate hybrid Cost-of-Pass proxy whose numerator is confirmation GPU cost per query and whose denominator is campaign MATH-500 pass@1; neither is Erol et al.'s per-problem estimator.

Table 1 places this measurement against those studies. The distinctive result here is estimand disagreement under a pinned stack, not a new compressor.

Table 1: Neighboring measurement studies. "Engine pinned" means one named serving stack is frozen while checkpoints vary. Length estimands and serving columns are what each paper reports as primary evidence, not an exhaustive inventory.

Study	Models	Bits	Engine pinned	Seeds	Length estimand	Correctness split	Serving / cost	Statistics
This paper	2 R1 distill (7B/8B)	BF16, FP8, AWQ-4, GPTQ-4	vLLM 0.7.0 eager, A100	5 MATH; 3 GSM/GPQA	RoM, paired Δ, BF16-correct Δ	2×2 + BF16 failure control	token proxy; seq. & batched GPU-s; $\widetilde{C}^{\mathrm{hyb}}_{\mathrm{pass}}$	clustered bootstrap, Holm, TOST
QRM [17]	R1 distill family	W/A/KV incl. W4A16	QRM harness (multi-config)	typically 1	mean tokens	not 2×2	limited	per-cell means
Lian et al. [13]	reasoning PTQ	low-bit PTQ	paper stack	paper-specific	BF16-correct Δ	BF16-correct vs rest	token cost	paired length
Lotfi et al. [18]	1.5B–32B reasoning	aggressive PTQ	paper stack	paper-specific	CoT length; overthinking	abandoned intermediates	logit-penalty Pareto	error-type counts
Kurtic et al. [10]	Llama-3.1 family	FP8/INT8/INT4	vLLM, several GPUs	large evals	not CoT-primary	N/A	sync vs continuous batching	accuracy + tok/s
Erol et al. [3]	general LMs	N/A	N/A	N/A	N/A	N/A	C_{pass} definition	economic
Hochlehnert et al. [9]	reasoning LMs	N/A	deliberately *un*pinned	many	N/A	N/A	N/A	reproducibility

3 Experimental Methodology

3.1 Models and precision formats

We evaluate DeepSeek-R1-Distill-Qwen-7B and DeepSeek-R1-Distill-Llama-8B [7]. Formats: uncompressed **BF16**; **FP8** checkpoints served on A100 via vLLM's Marlin **W8A16** weight-only fallback (not native W8A8 tensor-core FP8); **AWQ-4** (group size 128, `float16`); **GPTQ-4** (group size 128, Marlin). Hugging Face IDs and revisions are listed in the appendix. AWQ-4 checkpoints are third-party community uploads; accuracy losses on the tested Llama AWQ-4 checkpoint are properties of that artifact, not a proof about AWQ as a method.

3.2 Benchmarks and protocol

MATH-500 [8,14] ($n = 500$, seeds 42–46): 20,000 completions. GSM8K [2] ($n = 1{,}319$, seeds 42–44) and GPQA-Diamond [19] ($n = 198$, seeds 42–44): breadth. Decoding: zero-shot `<think>` prefix (appendix), $T = 0.6$, top-$p = 0.95$, repetition penalty 1.0 (deliberately not used to hide loops), max new tokens 32,768. Total: 56,408 completions in 88 checkpoint$\times$benchmark$\times$seed runs. MATH and GSM8K are widely used; a paired within-item design reduces but does not eliminate contamination interaction with format.

3.3 Hardware and serving stack

PARAM Rudra HPC (IIT Patna / NSM), NVIDIA A100-PCIE-80GB. Publication runs use the official QRM `inference.py` path [17] with `gpu_memory_utilization=0.75`. Engine pin: `vLLM==0.7.0`, eager execution. Source files under `configs/models/` are *not* the campaign launcher; they differ in `max_model_len` and KV-cache defaults and are not the effective campaign stack.

3.4 Measured serving

After the 56k accuracy campaign we timed the same eight checkpoints on A100-80GB under vLLM 0.7.0 eager. Primary numbers are the *controlled confirmation* (protocol: `docs/MEASURED_SERVING_CONFIRMATION_PROTOCOL.md`; raw JSON under `results/measured_serving_confirmation/raw/`). Condition B is the primary deployment-style measured serving condition; Condition A provides a sequential comparator, and the shared 65 tok/s token calculation is retained only as a historical sensitivity proxy. Condition A uses a balanced 20-prompt MATH-500 subset (4 per difficulty level), sequential single-request decode. Condition B uses a balanced 100-prompt subset (20 per level) in one `llm.generate` call with `max_num_seqs=8` pinned. Conditions A and B differ jointly in evaluated prompt subset and serving regime (20 sequential prompts versus 100 continuously batched prompts). Comparisons between them therefore measure sensitivity to the complete serving condition; they do not isolate batching or concurrency as the causal source of a ranking change. Secondary microbenchmark: 10 prompts forced to exactly 512 tokens (`ignore_eos`). Three warmup queries are discarded. Sampling matches the campaign ($T = 0.6$, top-$p = 0.95$) with seed 20260816. Repetitions share that seed. Aggregate output-token counts are identical across repetitions; therefore the observed throughput spread reflects runtime variation conditional on the same aggregate output-token count. The root cause of the observed slow/mid/fast timing regimes was not identified. The runner starts at $R = 3$ and expands to $R = 5$ when the *population* CV of the first three tok/s values exceeds 3%; this rule allocates two additional technical timing repetitions to cells with unusually high initial wall-clock variation and is a measurement-budget rule rather than an inferential stopping rule for model quality. Reported $\pm$ is *sample* SD. Qwen-7B jobs ran on `ragpu003` and Llama-8B jobs on `ragpu004` (one visible A100 each; PyTorch 2.5.1+cu124). Cross-architecture absolute tok/s is therefore cautious. An earlier unconstrained timing (`docs/MEASURED_SERVING_PROTOCOL.md`; mixed nodes; `max_num_seqs` not pinned) is not averaged with these numbers and is not used in the main tables.

Output tok/s = $(\sum \text{output tokens})/\Delta t_{\text{steady}}$. GPU-sec/query = $\Delta t_{\text{steady}}/N$ on one GPU. Peak VRAM is `torch.cuda.max_memory_allocated` after the 0.75 utilization preallocation ($\sim 60\,\text{GB}$ reserved of $80\,\text{GB}$). Measured

$$\widetilde{C}^{\text{hyb}}_{\text{pass}} = \frac{(\text{GPU-sec/query}) \times (\$1.50/3600)}{\text{pass@1}}, \tag{1}$$

using the frozen 40-cell MATH-500 means—not accuracy on the 100-item subset. We report an aggregate hybrid Cost-of-Pass proxy, $\widetilde{C}^{\text{hyb}}_{\text{pass}}$. Its numerator is mean confirmation GPU cost per query on the serving subset and its denominator is aggregate campaign MATH-500 pass@1. This aggregate ratio is inspired by Cost-of-Pass but is not the per-problem estimator defined by Erol et al. [3]. Serving prompts are not re-scored. Uncertainty intervals are 95% intervals conditional on the observed timing repetitions and a problem-clustered bootstrap of campaign pass@1 ($B = 10{,}000$). The \$1.50/A100-hour rate is a cloud pricing *scenario*, not a billed cluster price.

3.5 Metrics

Pass@1. Mean extractive match across seeds. Primary inference is a problem-clustered bootstrap of the paired difference $\Delta = \overline{\text{acc}}_{\text{quant}} - \overline{\text{acc}}_{\text{BF16}}$. For each problem i we first average correctness over seeds, $\delta_i = \overline{\text{acc}}_{i,\text{quant}} - \overline{\text{acc}}_{i,\text{BF16}}$, so all seeds belonging to a problem are retained in that item mean and treated as fixed rather than independently resampled. We then resample n problems with replacement ($B = 10{,}000$; RNG seed 0). The 95% intervals are percentile bootstrap intervals. Accordingly, these intervals quantify variation over benchmark problems conditional on the evaluated decoding-seed set and decoding configuration; decoding seeds are not treated as an additional sampled random effect. With bootstrap replicates Δ^*_b, the two-sided bootstrap tail-area p-value is

$$p_{\text{hi}} = \#\{\Delta^*_b \geq 0\}/B, \quad p_{\text{lo}} = \#\{\Delta^*_b \leq 0\}/B, \quad p = \min\bigl(1,\, 2\min(p_{\text{hi}}, p_{\text{lo}})\bigr). \tag{2}$$

A reported $p = 0$ or $p < 0.001$ means that no sign-crossing bootstrap replicate was observed among the 10,000 draws; it is not a literal probability of zero. Holm–Bonferroni is applied separately to the six primary within-benchmark format contrasts (MATH-500, GSM8K, GPQA-Diamond). Each benchmark is a distinct inferential family because each corresponds to a separate task distribution; we do not pool the 18 contrasts for the primary analysis. A joint Holm correction over all 18 contrasts is reported only as a stricter sensitivity (Appendix Table 17). Two one-sided tests (TOST) at a ± 1 pp equivalence margin use the 90% CI; this is an equivalence check, not a default claim. We use ± 1 pp as a deliberately narrow practical sensitivity margin for this study: a one-percentage-point benchmark-accuracy difference is treated as operationally small for this comparison. The margin is a study-level tolerance, not a universal equivalence standard for reasoning benchmarks.

maj@5 McNemar (secondary). A problem is maj@5-correct if at least 3 of 5 MATH-500 seeds match gold. Exact McNemar tests this consensus estimand, not pass@1. Non-significant maj@5 tests do not imply that pass@1 effects are absent.

Pathology. A row is a *loop* if the longest consecutive identical-word run is ≥ 20 (Unicode word tokens, case-folded). A row is a *cap hit* if re-encoded completion length is $\geq$ 32,768. Because decode-then-re-tokenize undershoots the cap (observed max 32,737), we also count *near-cap* rows with $\geq$ 32,500 tokens. Saved rows do not include vLLM `finish_reason` or output token IDs; phrase-level cycles are invisible to the identical-word detector.

Length. Primary estimator: ratio of means over all seeds, plus the mean paired per-item token delta with a clustered percentile bootstrap CI. We also report the mean of per-item ratios, which is a different estimand and is dominated by short BF16 traces. Length is stratified by the 2×2 of BF16/quantized extractive correctness, with clustered 95% CIs on each cell. A direct clustered contrast $D = \overline{\Delta}_{\text{BF16-only}} - \overline{\Delta}_{\text{Both-OK}}$ uses the same problem-level resamples for both stratum means; replicates in which either stratum is empty are skipped. D compares conditional means and is not a causal effect. A BF16-internal control compares BF16 length when BF16 is correct versus incorrect. The BF16-correct conditional Δ, following Lian et al., is the mean quantized−BF16 Δ on (item, seed) pairs where BF16 is correct (Both-OK $\cup$ BF16-only). The Both-OK interval assesses whether jointly-correct traces show a detectable length change, whereas D assesses whether mismatch-associated lengthening exceeds jointly-correct lengthening. A D interval entirely above zero supports that conditional difference. The mismatch strata condition on which member of the pair is wrong. Because failure traces are themselves long, the large and oppositely signed BF16-only and quantized-only deltas are partly a consequence of this conditioning. We therefore use D as a diagnostic of correctness-conditioned mismatch asymmetry, not as an estimate of a causal quantization-induced token increase. We do not interpret a long BF16-failure tail as a Liu–Lian reconciliation.

Historical token-implied aggregate cost proxy. Following the Cost-of-Pass ratio form of Erol et al. [3], a historical token-implied aggregate cost proxy is

$$\text{Cost per query} = (\bar{T}/\tau) \times (R/3600), \tag{3}$$

divided by campaign pass@1, with $\tau = 65$ tok/s (unmeasured) and $R = \$1.50$ per A100-hour. Because τ and R are shared, ranking by this proxy is ranking by $\bar{T}$/pass@1. It is an aggregate sensitivity proxy, not a measured per-problem Cost-of-Pass quantity. We retain it only as a sensitivity (Appendix Table 16); primary systems/cost evidence is $\widetilde{C}^{\text{hyb}}_{\text{pass}}$ in Section 3.4 and Table 11.

Observable modal agreement. Compact per-cell JSON stores `extractive_match` but not extracted strings. After recovering the 20,000 MATH-500 answer strings, we form agreement *without* gold. Campaign extraction and clustering used LightEval 0.8.0 with math-verify (frozen in `docs/ANSWER_NORMALIZATION.md`). A later throwaway MacBook install of LightEval 0.8.1 was *not* used for canonical numbers. For item i and five generated predictions, let $\sim$ be that campaign mathematical-equivalence relation. Cluster the five predictions into connected components of $\sim$. Let m_i be the size of the unique largest class when one exists; otherwise the item is a non-unique-mode tie. Observable agreement is $a_i = m_i/5$. Serve at threshold $k/5$ only when a unique modal class satisfies $m_i \geq k$. The pattern (A, A, B, B, C) has $m_i = 2$ and is not served at $\geq 3/5$; (A, A, A, B, B) has unique mode A with $m_i = 3$ and is served at $\geq 3/5$. Gold is used only afterward to estimate selective accuracy and risk. All five samples are generated before the decision. The five-sample token-cost proxy T_5 is the sum of their completion tokens, including

Table 2: MATH-500 headline grid ($n = 500$, seeds 42–46). Clustered 95% CIs resample problems. Loops: identical-word run ≥ 20. Near-cap: completion tokens $\geq 32{,}500$. Exact cap hits after re-encoding: 0 in all cells.

Cell	**42**	**43**	**44**	**45**	**46**	**Mean ± Std**	**Clust. 95%**	**Tok.**	**L**	**NC**
Qwen BF16	94.4	94.0	93.8	94.6	93.2	94.00 ± 0.55	[92.2, 95.6]	4,011.4	0	14
Qwen FP8	94.4	95.2	94.8	92.6	95.0	94.40 ± 1.05	[92.8, 96.0]	4,007.7	0	15
Qwen AWQ-4	92.4	92.8	93.2	93.0	94.2	93.12 ± 0.67	[91.4, 94.8]	4,265.3	1	25
Qwen GPTQ-4	93.8	92.6	93.4	94.6	93.0	93.48 ± 0.77	[91.7, 95.2]	4,287.2	0	24
Llama BF16	89.0	88.4	90.2	89.8	88.8	89.24 ± 0.74	[87.2, 91.2]	4,656.6	3	19
Llama FP8	89.0	89.6	88.6	89.2	91.2	89.52 ± 1.01	[87.4, 91.6]	4,550.8	3	10
Llama AWQ-4	84.4	84.8	89.2	87.4	86.6	86.48 ± 1.96	[84.2, 88.7]	4,736.5	3	12
Llama GPTQ-4	88.0	89.6	86.8	89.4	90.8	88.92 ± 1.55	[86.9, 90.9]	4,840.5	2	21

Table 3: Primary: problem-clustered bootstrap of MATH-500 pass@1 versus BF16 ($B = 10{,}000$). Secondary: exact McNemar on maj@5. Holm at $\alpha = 0.05$ over six contrasts (primary family). TOST ± 1 pp uses the 90% CI. sig.: Holm-significant at $\alpha = 0.05$; n.s.: not significant.

Contrast	Δ (pp)	**95% CI**	p	**Holm**	**90% CI**	**TOST**	**McNemar** p
Llama AWQ-4	-2.76	$[-4.16, -1.44]$	< 0.001	sig.	$[-3.92, -1.64]$	fail	0.296
Qwen AWQ-4	-0.88	$[-1.88, +0.08]$	0.069	n.s.	$[-1.68, -0.08]$	fail	1.000
Qwen GPTQ-4	-0.52	$[-1.40, +0.36]$	0.245	n.s.	$[-1.28, +0.20]$	fail	0.754
Qwen FP8	$+0.40$	$[-0.48, +1.28]$	0.383	n.s.	$[-0.32, +1.12]$	fail	0.581
Llama GPTQ-4	-0.32	$[-1.56, +0.96]$	0.619	n.s.	$[-1.36, +0.72]$	fail	1.000
Llama FP8	$+0.28$	$[-0.92, +1.48]$	0.647	n.s.	$[-0.72, +1.28]$	fail	1.000

abstained queries. We report three discrete operating points, not a continuous risk–coverage curve. Gold-hit ECE (confidence $= c_i/5$, label $= \mathbb{I}(c_i \geq 3)$) remains a deterministic function of the gold-hit histogram and is omitted.

4 Results

4.1 MATH-500 pass@1

Table 2 and Figure 5 show per-seed pass@1. FP8 point estimates sit 0.28–0.40 pp above BF16. The tested Llama AWQ-4 checkpoint shows the largest MATH-500 pass@1 drop among the evaluated Llama cells (-2.76 pp).

Table 3 is the test that matches RQ1. The tested Llama AWQ-4 checkpoint's clustered interval excludes zero ($p < 0.001$; Holm-significant among the six MATH contrasts). Qwen 4-bit MATH drops are negative but not Holm-significant. No MATH contrast, including FP8, has its 90% CI inside $[-1, +1]$ pp, so we do *not* claim statistical equivalence at a 1 pp serving margin. Secondary maj@5 McNemar p-values are all > 0.29: that consensus estimand does not capture the per-seed flips that pass@1 still sees.

4.2 GSM8K and GPQA-Diamond

Table 8 gives cell means; Tables 4 and 5 are the clustered tests. FP8 stays within about one point of BF16 on all three tasks; several GSM8K Qwen (and Llama FP8/GPTQ) contrasts pass TOST at ± 1 pp. The tested Qwen AWQ-4 checkpoint decreases GPQA-Diamond pass@1 by 5.56 pp (95% CI $[-9.60, -1.52]$, $p = 0.007$; 3 seeds, $n = 198$) and is Holm-significant within the primary six-contrast GPQA family; it does not remain significant under the stricter exploratory

Table 4: GSM8K clustered pass@1 versus matched BF16 (n = 1,319, 3 seeds). Holm over six contrasts. TOST ±1 pp uses the 90% CI. sig.: Holm-significant at $\alpha = 0.05$; n.s.: not significant.

Contrast	Δ (pp)	**95% CI**	p	**Holm**	**90% CI**	**TOST**
Llama AWQ-4	−1.57	[−2.55, −0.58]	0.002	sig.	[−2.38, −0.73]	fail
Qwen AWQ-4	−0.20	[−0.91, +0.48]	0.579	n.s.	[−0.78, +0.38]	pass
Qwen GPTQ-4	−0.13	[−0.86, +0.63]	0.729	n.s.	[−0.76, +0.51]	pass
Llama GPTQ-4	+0.28	[−0.53, +1.09]	0.511	n.s.	[−0.40, +0.96]	pass
Qwen FP8	+0.08	[−0.58, +0.73]	0.827	n.s.	[−0.45, +0.63]	pass
Llama FP8	+0.13	[−0.66, +0.94]	0.761	n.s.	[−0.53, +0.81]	pass

Table 5: GPQA-Diamond clustered pass@1 versus matched BF16 (n = 198, 3 seeds). Holm over six contrasts. sig.: Holm-significant at $\alpha = 0.05$; n.s.: not significant. The Qwen AWQ-4 row is the largest accuracy effect in the grid.

Contrast	Δ (pp)	**95% CI**	p	**Holm**	**90% CI**	**TOST**
Qwen AWQ-4	−5.56	[−9.60, −1.52]	0.007	sig.	[−9.09, −2.19]	fail
Qwen GPTQ-4	−2.36	[−6.57, +1.85]	0.288	n.s.	[−6.06, +1.18]	fail
Llama GPTQ-4	−1.18	[−5.39, +3.03]	0.573	n.s.	[−4.71, +2.36]	fail
Qwen FP8	−0.84	[−4.71, +3.03]	0.690	n.s.	[−4.21, +2.53]	fail
Llama AWQ-4	+0.84	[−3.54, +5.05]	0.711	n.s.	[−2.86, +4.38]	fail
Llama FP8	+1.68	[−2.19, +5.39]	0.403	n.s.	[−1.52, +4.88]	fail

Holm correction across all 18 benchmark contrasts (Appendix Table 17). The tested Llama AWQ-4 checkpoint also drops on GSM8K (−1.57 pp, 95% CI [−2.55, −0.58], $p = 0.002$). GPQA intervals are otherwise wide; we treat non-AWQ GPQA contrasts as exploratory.

4.3 Loops and near-cap completions

Across 88 checkpoint×benchmark×seed runs the validator records **25** repetition-flagged rows (0.044% of 56,408) and **0** exact cap hits. Loops occur in BF16 as well as quantized cells (Llama MATH: 3/3/3/2 for BF16/FP8/AWQ/GPTQ). Three runs exceed 10,000 identical consecutive words. Near-cap completions ($\geq$ 32,500 tokens) total 209; on MATH-500, Qwen AWQ-4/GPTQ-4 have 25 and 24 versus BF16's 14. No row re-encoded to $\geq$ 32,768; native stop reasons were not logged.

4.4 Token length

On the *full* MATH-500 grid and all five seeds (Table 6, Figure 3), Qwen AWQ-4/GPTQ-4 generate +6.33%/+6.88% more tokens than BF16 (mean paired Δ +254 and +276 tokens; CIs exclude zero). Excluding near-cap pairs ($\geq$ 32,500 tokens) the ratios remain +4.61%/+5.75%. Llama GPTQ-4 is +3.95% (+184 tokens, CI just excludes zero). Llama AWQ-4 +1.72% and both FP8 cells have CIs that include zero.

A 200-item even-index, seed-42, mean-of-ratios analysis is a superseded estimator (Appendix G); it is not the full-grid ratio of means.

Figure 4 and Table 6 split paired token deltas by the 2×2. The primary descriptive length evidence is the full-grid ratio of means, the paired mean token Δ, and the Both-OK Δ among jointly correct pairs. Qwen AWQ-4 and GPTQ-4 Both-OK CIs exclude zero (+166 [+80, +254] and +136 [+45, +239]), so jointly-correct traces still lengthen on those cells. The mismatch strata (BF16-only / Quant-only), mismatch-excess D, and the BF16 failure-length control are diagnostics. The large, oppositely signed mismatch-stratum deltas show that correctness-conditioned length summaries are strongly influenced by the long failure tail of whichever check-

Table 6: MATH-500 length versus BF16, all five seeds. RoM: ratio of means. Δ: mean paired per-item token difference with clustered 95% CI. 2×2 cells are mean quantized$-$BF16 Δ with clustered 95% CI and pair count n. BF16-correct column: BF16-correct conditional Δ, following Lian et al. BF16 len.: mean BF16 tokens when BF16 is correct vs. incorrect.

Contrast	RoM	Mean Δ [95% CI]	Both OK	BF16 only	Quant only	Both wrong	BF16-correct Δ [95% CI]
Qwen FP8	-0.1%	-4 $[-142, +134]$	$+51$ $[-42, +145]$ (n=2297)	$+5444$ $[+2922, +8035]$ (n=53)	-6940 $[-9479, -4345]$ (n=63)	$+264$ $[-1126, +1685]$ (n=87)	$+172$ $[+63, +291]$
Qwen AWQ-4	$+6.3\%$	$+254$ $[+116, +398]$	$+166$ $[+80, +254]$ (n=2272)	$+6915$ $[+4534, +9345]$ (n=78)	-6533 $[-8933, -4255]$ (n=56)	$+894$ $[-620, +2420]$ (n=94)	$+390$ $[+261, +531]$
Qwen GPTQ-4	$+6.9\%$	$+276$ $[+121, +431]$	$+136$ $[+45, +239]$ (n=2284)	$+6892$ $[+4090, +9896]$ (n=66)	-4910 $[-8180, -1715]$ (n=53)	$+1893$ $[+204, +3388]$ (n=97)	$+326$ $[+198, +464]$
Llama FP8	-2.3%	-106 $[-284, +65]$	$+38$ $[-74, +152]$ (n=2121)	$+2629$ $[+1098, +4265]$ (n=110)	-4159 $[-6040, -2428]$ (n=117)	-968 $[-2252, +198]$ (n=152)	$+166$ $[+33, +302]$
Llama AWQ-4	$+1.7\%$	$+80$ $[-109, +268]$	$+91$ $[-25, +210]$ (n=2052)	$+3090$ $[+1995, +4286]$ (n=179)	-4477 $[-6503, -2588]$ (n=110)	-305 $[-1520, +907]$ (n=159)	$+332$ $[+182, +490]$
Llama GPTQ-4	$+3.9\%$	$+184$ $[+12, +359]$	$+66$ $[-43, +181]$ (n=2097)	$+5915$ $[+4335, +7605]$ (n=134)	-4327 $[-6138, -2561]$ (n=126)	$+521$ $[-506, +1585]$ (n=143)	$+417$ $[+261, +583]$

Table 7: Clustered mismatch excess $D = \overline{\Delta}_{\mathrm{BF16-only}} - \overline{\Delta}_{\mathrm{Both-OK}}$ on MATH-500. Both stratum means are formed on the same problem-level resamples ($B = 10{,}000$; empty-stratum replicates skipped; none occurred). D is a diagnostic of correctness-conditioned mismatch asymmetry, not a causal quantization-induced token increase.

Contrast	Both-OK Δ	BF16-only Δ	Mismatch excess D [95% CI]
Qwen FP8	$+51$ $[-42, +145]$	$+5444$ $[+2922, +8035]$	$+5394$ $[+2853, +7992]$
Qwen AWQ-4	$+166$ $[+80, +254]$	$+6915$ $[+4534, +9345]$	$+6748$ $[+4375, +9184]$
Qwen GPTQ-4	$+136$ $[+45, +239]$	$+6892$ $[+4090, +9896]$	$+6755$ $[+3938, +9786]$
Llama FP8	$+38$ $[-74, +152]$	$+2629$ $[+1098, +4265]$	$+2591$ $[+1047, +4244]$
Llama AWQ-4	$+91$ $[-25, +210]$	$+3090$ $[+1995, +4286]$	$+2999$ $[+1898, +4187]$
Llama GPTQ-4	$+66$ $[-43, +181]$	$+5915$ $[+4335, +7605]$	$+5849$ $[+4258, +7553]$

point is wrong. D therefore serves as a mismatch diagnostic (Table 7); it is not an estimate of a causal quantization-induced token increase. The Both-OK contrast gives the cleaner descriptive comparison among jointly correct outputs and is itself not causal. Qwen FP8 and all three Llama Both-OK CIs include zero. The BF16-correct conditional Δ, following Lian et al., is positive with CIs excluding zero in every contrast (+166 to +417 tokens). BF16-incorrect traces are themselves long: 15,843 vs. 3,256 tokens on Qwen and 13,928 vs. 3,539 on Llama. We report that control; we do not call it a Liu–Lian reconciliation.

Under the shared-τ proxy (Appendix Table 16), FP8 ranks first among Qwen cells. Measured Conditions A and B disagree with that order (Section 4.7).

4.5 Difficulty labels

Table 9: Level 1 is easy for every format in this sample. The tested Llama AWQ-4 checkpoint is weakest on Levels 2–4. MATH level labels are not a monotone difficulty axis here (Level 2 Qwen BF16 89.11% vs Level 5 92.69%).

4.6 Observable modal agreement on MATH-500

Because MATH-500 was evaluated with five generations per item, we additionally examine whether observable answer agreement provides a useful gold-free abstention signal. Recovered answer strings make RQ3 measurable as observable multi-sample agreement, selective prediction, and abstention—not as model safety or softmax calibration, and not as G-Pass@k [16]. Table 10 reports coverage and selective risk at three discrete thresholds. On MATH-500, higher observable answer agreement is associated with lower selective error. At 5/5 consensus, observed selective error is at most 0.27% across the evaluated configurations, but coverage ranges from 70.2% (Llama AWQ-4) to 88.8% (Qwen FP8). Several 0.00% risk cells have bootstrap intervals $[0, 0]$; Wilson 95% upper bounds on those $0/n$ cells are 0.82%–1.08%. Zero observed errors is not zero true error.

AWQ-4 reduces strict-consensus coverage in some configurations, most clearly for Llama-8B: 70.2% versus 76.2% for BF16, a paired difference of -6.0 pp (95% CI $[-9.4, -2.6]$). Llama

Table 8: Cross-benchmark pass@1 (mean ± seed std). MATH-500: 5 seeds. GSM8K and GPQA-Diamond: 3 seeds. L/NC: loops / near-cap over those seeds.

Cell	MATH-500	GSM8K	GPQA-D	L / NC
Qwen BF16	94.00 ± 0.55	91.26 ± 0.29	50.34 ± 2.96	4 / 28
Qwen FP8	94.40 ± 1.05	91.33 ± 0.16	49.49 ± 1.52	0 / 24
Qwen AWQ-4	93.12 ± 0.67	91.05 ± 1.14	44.78 ± 3.04	5 / 32
Qwen GPTQ-4	93.48 ± 0.77	91.13 ± 0.27	47.98 ± 1.75	3 / 40
Llama BF16	89.24 ± 0.74	88.68 ± 0.46	46.13 ± 1.91	3 / 21
Llama FP8	89.52 ± 1.01	88.80 ± 0.62	47.81 ± 0.29	3 / 12
Llama AWQ-4	86.48 ± 1.96	87.11 ± 0.23	46.97 ± 2.02	4 / 21
Llama GPTQ-4	88.92 ± 1.55	88.96 ± 0.70	44.95 ± 4.32	3 / 31

Table 9: MATH-500 difficulty labels (5-seed pass@1 mean ± std). Labels are not monotone: Level 2 is harder than Level 5 for both BF16 backbones. Level-1 $n = 43$; treat cell-wise gaps as descriptive.

Level	Qwen BF16	Qwen FP8	Qwen AWQ-4	Qwen GPTQ-4
1 ($n = 43$)	98.60 ± 1.27	98.60 ± 2.08	98.14 ± 1.95	96.28 ± 2.08
2 ($n = 90$)	89.11 ± 0.93	90.44 ± 4.20	90.67 ± 1.69	88.67 ± 2.14
3 ($n = 105$)	95.81 ± 1.09	95.24 ± 1.51	94.10 ± 0.80	94.29 ± 1.90
4 ($n = 128$)	95.78 ± 0.89	96.41 ± 1.42	94.53 ± 1.24	96.56 ± 1.18
5 ($n = 134$)	92.69 ± 1.11	93.13 ± 1.23	91.04 ± 1.67	92.24 ± 1.95
Level	**Llama BF16**	**Llama FP8**	**Llama AWQ-4**	**Llama GPTQ-4**
1 ($n = 43$)	93.95 ± 3.12	94.88 ± 3.03	95.35 ± 1.64	96.28 ± 2.65
2 ($n = 90$)	84.67 ± 2.14	85.11 ± 2.02	81.11 ± 3.04	83.33 ± 3.42
3 ($n = 105$)	89.90 ± 2.48	90.10 ± 1.73	85.71 ± 1.17	89.71 ± 1.83
4 ($n = 128$)	91.56 ± 1.69	91.09 ± 0.43	87.03 ± 2.25	89.84 ± 1.75
5 ($n = 134$)	88.06 ± 1.75	88.81 ± 2.84	87.31 ± 3.46	88.81 ± 1.18

AWQ-4 $\geq 4/5$ coverage falls 3.2 pp ($[-6.0, -0.6]$). Qwen AWQ-4 $\geq 4/5$ coverage falls 2.2 pp ($[-4.0, -0.6]$). The $\geq 3/5$ AWQ contrasts, and all GPTQ-4 coverage contrasts, have paired intervals that include zero. No clear FP8–BF16 difference was detected in modal-agreement coverage or selective risk under these configurations. Appendix Table 12 lists the full threshold pattern.

The five-sample token-cost proxy is paid before abstention. Mean T_5 is 20,057 / 20,038 / 21,327 / 21,436 tokens for Qwen BF16/FP8/AWQ-4/GPTQ-4 and 23,283 / 22,754 / 23,682 / 24,203 for Llama. Qwen 4-bit retains its approximately +6–7% output-token penalty in this accounting because T_5 is the sum of the same five traces. Tokens per served query divide that total by the number served, so abstained items are not free. This is not a measured monetary cost. Figure 1 shows the three operating points.

4.7 Measured serving throughput and Cost-of-Pass

RQ4 asks whether checkpoint rankings agree across the historical token proxy, sequential Condition A, and batched Condition B aggregate serving-cost proxies. They do not. Table 11 places Condition A and Condition B beside each other with $\widetilde{C}_{\text{pass}}^{\text{hyb}}$ 95% intervals conditional on the observed timing repetitions and clustered accuracy bootstrap. Among Qwen cells the 65 tok/s proxy ranks FP8 < BF16 < AWQ-4 < GPTQ-4; Condition A ranks AWQ-4 < GPTQ-4 < FP8 < BF16; Condition B ranks GPTQ-4 < AWQ-4 < FP8 < BF16 (lower $\widetilde{C}_{\text{pass}}^{\text{hyb}}$ first). Among Llama cells the proxy ranks FP8 first, Condition A ranks BF16 first, and Condition B ranks

Table 10: Observable modal-answer agreement on MATH-500 ($n = 500$, seeds 42–46). Coverage is the fraction of items with a unique modal class of size $\geq k$. Risk is 1−selective accuracy among served items. Mean T_5 is the five-sample token-cost proxy over all 500 items (abstained queries still pay). Gold is used only to score served predictions.

Model	**Fmt**	≥3/5 cov.	≥3/5 risk	≥4/5 cov.	≥4/5 risk	5/5 cov.	5/5 risk	Mean T_5
Qwen-7B	BF16	96.0	1.67	93.4	0.43	88.4	0.23	20,057
Qwen-7B	FP8	96.6	1.66	92.6	0.00	88.8	0.00	20,038
Qwen-7B	AWQ-4	95.8	1.46	91.2	0.44	86.4	0.23	21,327
Qwen-7B	GPTQ-4	95.4	1.47	92.8	0.43	86.8	0.00	21,436
Llama-8B	BF16	94.0	2.98	87.8	0.68	76.2	0.26	23,283
Llama-8B	FP8	94.0	3.19	88.6	0.90	78.2	0.00	22,754
Llama-8B	AWQ-4	93.2	3.65	84.6	1.89	70.2	0.00	23,682
Llama-8B	GPTQ-4	93.8	2.77	86.4	0.46	75.4	0.27	24,203

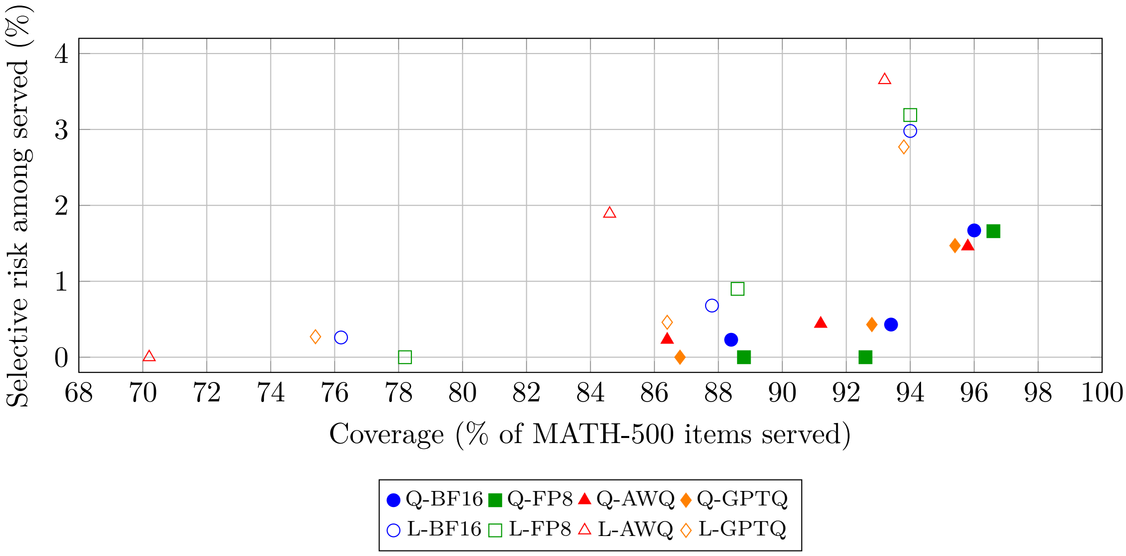

Figure 1: Three discrete MATH-500 operating points ($\geq 3/5$, $\geq 4/5$, 5/5), not a continuous risk–coverage curve. Filled markers: Qwen-7B. Open markers: Llama-8B. Lower-right is high coverage and higher residual risk; lower-left is strict consensus.

FP8 first. That disagreement is serving-condition sensitivity, not an isolated batching effect. No unique cheapest checkpoint is identified across these estimators.

Qwen GPTQ-4 Condition B is −45.9% vs. Qwen BF16 (95% CI [−46.4, −45.4]) at \$1.50/A100-hour. Qwen AWQ-4 Condition A is cheaper than Qwen GPTQ-4 (\$0.0215 vs. \$0.0242). Qwen FP8 Condition B mean $\widetilde{C}^{\text{hyb}}_{\text{pass}}$ is −36.0% vs. BF16, but the five repeats show slow/mid/fast runtime variation at identical aggregate output-token counts: two slow $\sim$ 351 tok/s ($\widetilde{C}^{\text{hyb}}_{\text{pass}}$ \$0.0047), one mid $\sim$ 456 tok/s (\$0.0036), two fast $\sim$ 545 tok/s (\$0.0030). We keep all five; Appendix Table 15 lists them. The root cause of those timing regimes was not identified. Llama-8B FP8 Condition B is −8.2% $\widetilde{C}^{\text{hyb}}_{\text{pass}}$ (95% CI [−9.5, −6.9]). Llama AWQ-4 and GPTQ-4 raise Condition B $\widetilde{C}^{\text{hyb}}_{\text{pass}}$ (+18.5% and +24.9%). Llama GPTQ-4 Condition B mean throughput is about 0.2% lower than Llama BF16 (366.16 vs. 366.98 tok/s). These FP8 numbers are A100 Marlin W8A16, not Hopper W8A8.

Four-bit speed is not one number. Confirmation Condition A: Qwen AWQ-4 and GPTQ-4 exceed matched BF16 tok/s (76.89 and 82.67 vs. 43.92), whereas the tested Llama AWQ-4 and GPTQ-4 checkpoints do not (70.20 and 63.49 vs. 72.34). Batched Qwen GPTQ-4 reaches 488.26 ± 0.64 tok/s. Serving effects are checkpoint- and serving-condition-specific under this

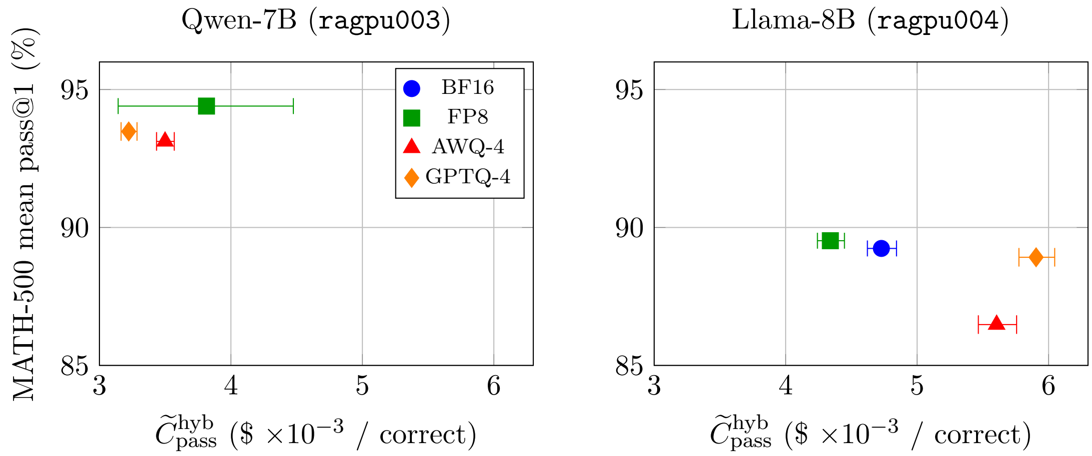

Figure 2: Confirmation Condition B aggregate hybrid Cost-of-Pass proxy versus campaign MATH-500 pass@1 (\$ $\times 10^{-3}$). Whiskers are 95% uncertainty intervals conditional on the observed timing repetitions and clustered accuracy bootstrap. Points are not joined. Panels are interpreted within model family; Qwen and Llama confirmation jobs ran on different hosts, so no pooled cross-host frontier is inferred. Within Qwen, Condition B nondominated points are FP8 and GPTQ-4; within Llama, FP8. Not a unique Pareto-efficient cell. Qwen FP8's wide whisker is the five-rep wall-clock spread.

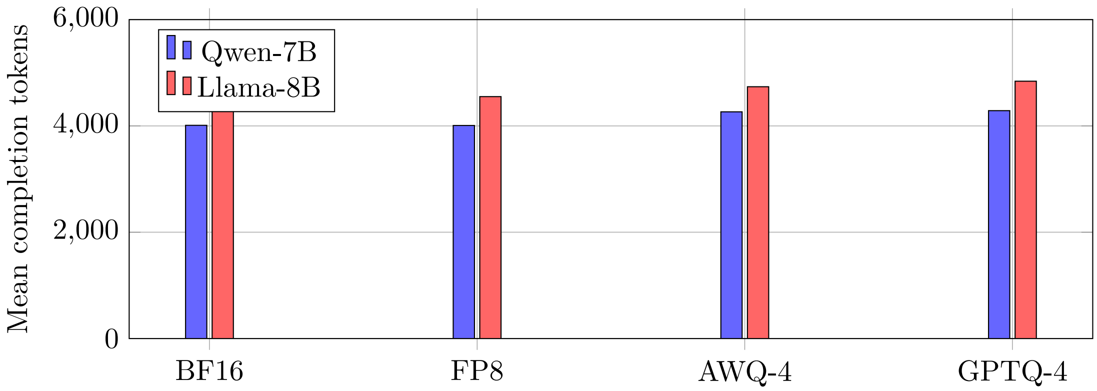

Figure 3: Full-grid mean MATH-500 completion tokens (ratio-of-means estimand).

implementation.

Token inflation does not automatically cancel a kernel speedup. Cost is GPU-seconds $\times$ a pricing scenario, not tokens alone. No serving-confirmation *mean* output reached the 32,768-token generation cap. Compact confirmation JSON does not store native `finish_reason`.

Peak allocated VRAM is 53.9–56.1 GB for every batched confirmation cell. That is the 0.75 utilization pool, not a measurement of 4-bit weight-footprint savings. We do not publish a format-to-format weight table from `torch.cuda.memory_allocated` after engine init.

On (pass@1, $\widetilde{C}^{\mathrm{hyb}}_{\mathrm{pass}}$) within Qwen on Condition B, the nondominated points are Qwen-7B FP8 and Qwen-7B GPTQ-4. Within Llama, the only nondominated Condition B point is Llama FP8. Because Qwen and Llama confirmation jobs ran on different hosts, we do not infer a pooled cross-host frontier (Figure 2). Latency detail for Condition A remains in Appendix Table 13.

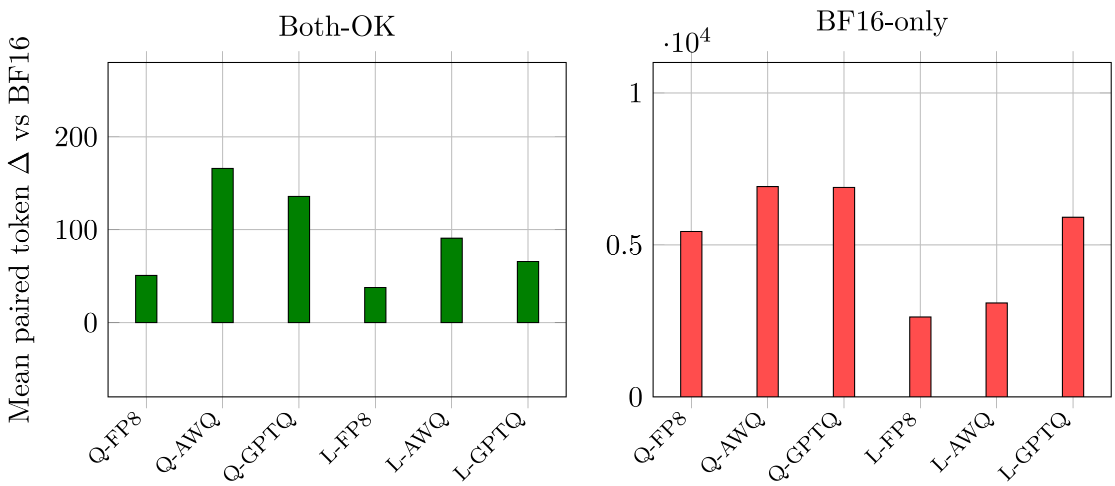

Figure 4: Paired token deltas by correctness, two panels so Both-OK is visible. Left: jointly-correct pairs (Qwen AWQ/GPTQ CIs exclude 0; FP8 and Llama CIs include 0; n in Table 6). Right: BF16-correct / quantized-wrong pairs. Clustered 95% CIs and n are in Table 6.

Table 11: Aggregate hybrid Cost-of-Pass proxy $\widetilde{C}^{\text{hyb}}_{\text{pass}}$ on the controlled confirmation. Condition A: 20 sequential prompts. Condition B: 100-prompt `llm.generate`, `max_num_seqs=8`. Tok/s is mean± sample SD. The numerator is confirmation GPU-sec/query; the denominator is campaign MATH-500 pass@1 at \$1.50/A100-hour. Intervals are 95% (timing-rep × clustered pass@1), conditional on the observed timing repetitions. Qwen: `ragpu003`. Llama: `ragpu004`.

Model	Fmt	A tok/s	A GPU-s/q	A $\widetilde{C}^{\text{hyb}}_{\text{pass}}$ [95% CI]	B tok/s	B GPU-s/q	B $\widetilde{C}^{\text{hyb}}_{\text{pass}}$ [95% CI]	B Δ% vs BF16
Qwen-7B	BF16	43.92 ± 0.04	102.51	\$0.0454 [0.0447, 0.0463]	252.72 ± 0.56	13.44	\$0.0060 [0.0059, 0.0061]	anchor
Qwen-7B	FP8	62.50 ± 0.08	84.60	\$0.0373 [0.0367, 0.0380]	449.79 ± 97.53	8.64	\$0.0038 [0.0031, 0.0045]	−36.0 [−47.2, −24.8]
Qwen-7B	AWQ-4	76.89 ± 0.70	48.04	\$0.0215 [0.0211, 0.0219]	418.15 ± 2.03	7.82	\$0.0035 [0.0034, 0.0036]	−41.3 [−41.9, −40.6]
Qwen-7B	GPTQ-4	82.67 ± 0.69	54.39	\$0.0242 [0.0238, 0.0248]	488.26 ± 0.64	7.23	\$0.0032 [0.0032, 0.0033]	−45.9 [−46.4, −45.4]
Llama-8B	BF16	72.34 ± 0.27	66.43	\$0.0310 [0.0303, 0.0318]	366.98 ± 1.90	10.13	\$0.0047 [0.0046, 0.0048]	anchor
Llama-8B	FP8	78.75 ± 2.89	92.55	\$0.0431 [0.0415, 0.0450]	481.42 ± 1.36	9.32	\$0.0043 [0.0042, 0.0044]	−8.2 [−9.5, −6.9]
Llama-8B	AWQ-4	70.20 ± 2.71	89.51	\$0.0431 [0.0415, 0.0450]	391.11 ± 0.48	11.63	\$0.0056 [0.0055, 0.0058]	+18.5 [+16.6, +20.5]
Llama-8B	GPTQ-4	63.49 ± 0.41	86.95	\$0.0407 [0.0398, 0.0418]	366.16 ± 0.79	12.60	\$0.0059 [0.0058, 0.0060]	+24.9 [+23.0, +26.7]

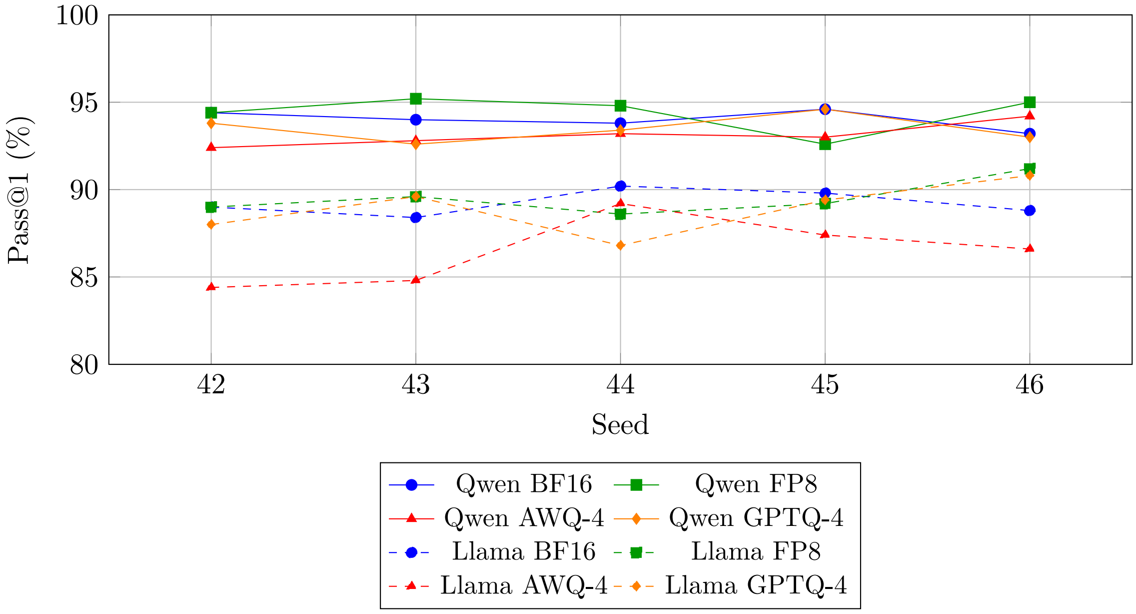

Figure 5: MATH-500 pass@1 by seed. Solid: Qwen-7B. Dashed: Llama-8B. The tested Llama AWQ-4 checkpoint shows the largest MATH-500 pass@1 drop among the evaluated Llama cells.

5 Discussion

Under this pinned A100 W8A16 stack, evaluating these eight public checkpoints requires more than a single pass@1 or token count. FP8–BF16 MATH intervals include zero, but TOST at ± 1 pp fails, so we do not treat FP8 as a statistically equivalent accuracy default. Four-bit behavior is checkpoint- and task-specific: the tested Qwen AWQ-4 checkpoint keeps MATH-500 and GSM8K near BF16 while losing 5.56 pp on GPQA-Diamond (Holm-significant within the primary six GPQA contrasts; not under the exploratory 18-contrast Holm correction); the tested Llama AWQ-4 checkpoint shows the largest MATH-500 and GSM8K pass@1 drops among the evaluated Llama cells. Mean Qwen 4-bit MATH-500 length rises 6.3–6.9%. Both-OK CIs exclude zero for those Qwen 4-bit cells, so jointly-correct traces still lengthen. Mismatch-conditioned differences are much larger because failure traces are themselves long; D is a diagnostic of that asymmetry, not a causal compression effect.

Pass@1 and maj@5 are different estimands. Llama AWQ-4's degradation appears in both mean accuracy and 5/5 consensus coverage. Qwen 4-bit formats retain relatively strong MATH-500 pass@1 while spending more tokens. No clear FP8–BF16 difference was detected in modal-agreement coverage or selective risk under these configurations; paired intervals that include zero do not prove equivalence. Repeated sampling sharply reduces selective risk on MATH-500 but costs about five generations per attempted item. Wilson intervals on $0/n$ cells remain strictly positive. The construction is a gold-free unique-mode abstention rule on five MATH seeds, not G-Pass@k and not a safety property.

A 200-item even-index length estimator, an unconstrained serving timing, and a vLLM 0.8.5 pathology autopsy are superseded analyses (Appendix G); they are not mixed with the canonical 56k grid or the confirmation serving numbers.

Cost estimators disagree. The historical 65 tok/s token proxy, sequential Condition A, and batched Condition B rank the four Qwen (resp. Llama) cells differently. Qwen GPTQ-4 Condition B $\widetilde{C}_{\text{pass}}^{\text{hyb}}$ is -45.9% vs. BF16, but Condition A ranks Qwen AWQ-4 first, and Qwen FP8 Condition B is a five-rep mixture of slow and fast wall-clock regimes whose cause was not identified. That A/B disagreement is serving-condition sensitivity (workload and serving regime jointly), not an isolated batching effect. Llama GPTQ-4 Condition B mean throughput is about 0.2% lower than Llama BF16. Peak allocated VRAM did not separate formats under the 0.75 utilization cap and reflects the engine allocation pool rather than isolated weight-footprint compression.

The deployment implication is not that one quantization format is universally preferable. Public checkpoints should be evaluated on the target task and intended serving condition: token-based proxies can mis-rank measured GPU cost, sequential and continuously batched workloads can reorder checkpoint rankings, and community quantization artifacts can show task-specific accuracy losses. Deployment decisions should therefore combine checkpoint-specific accuracy with direct measurements from the intended serving configuration.

6 Limitations

1. **A100 W8A16 fallback.** FP8 numbers do not transfer to Hopper native W8A8.
2. **Measured serving scope.** Throughput, latency, GPU-sec/query, and $\widetilde{C}_{\text{pass}}^{\text{hyb}}$ are measured on balanced MATH-500 serving subsets (20 sequential prompts for Condition A; 100 batched prompts for Condition B), not the full 56k campaign. Pass@1 in the proxy denominator is the 40-cell campaign mean. Both serving conditions use difficulty-balanced subsets rather than the natural MATH-500 level distribution. Conditions A and B also differ jointly in prompt subset and serving regime. Because continuous-batch Condition B records aggregate elapsed time rather than per-request or per-level service time, the retained data cannot be retrospectively reweighted to identify natural-distribution GPU

cost, and the A/B ranking reversal should be interpreted as serving-condition sensitivity rather than a causal batching effect. The \$1.50/A100-hour rate is a pricing scenario, not billed cluster cost. Confirmation Condition B pins `max_num_seqs=8`. Repetitions share sampling seed 20260816. Qwen and Llama confirmation jobs used different hosts (`ragpu003` vs. `ragpu004`); within-architecture comparisons are same-node. Qwen FP8 Condition B retains five wall-clock repeats with large timing variance at identical aggregate output-token counts; the cause was not identified. Peak allocated VRAM is the 0.75 engine pool, not isolated weight footprint. Compact confirmation JSON has no `finish_reason`. Energy is not measured.

3. **Pathology metrology.** The loop detector is identical-word runs ≥ 20. Phrase-level cycles and `finish_reason` were not saved in the original campaign. Near-cap is a proxy.

4. **Modal agreement.** Five generations are required per attempted MATH problem. Only MATH-500 has five seeds in this campaign. Mathematical answer equivalence is evaluator- and task-specific (campaign LightEval 0.8.0 / math-verify). High agreement can still be wrong. The signal is not general uncertainty calibration, not softmax calibration, and not a safety property.

5. **Checkpoint provenance.** AWQ-4 is a community quantization of unknown calibration set and quantizer revision. GPTQ-4/FP8 are RedHatAI artifacts (appendix).

6. **Scope.** Two 7–8B DeepSeek-R1 distillations, English math/science, one pinned serving stack, one prompt. GSM8K/GPQA use three seeds.

7. **Contamination.** MATH and GSM8K overlap public pretraining corpora. A paired within-item design reduces but does not eliminate contamination interaction with checkpoint.

8. **Source configs vs. launcher.** Files in `configs/models/` are not the QRM `inference.py` campaign. Effective KV-cache dtype is not stored per cell. Launcher flags such as `max_model_len=32768` are code/launcher-confirmed, not per-job runtime telemetry.

7 Conclusion

Under a pinned A100 / vLLM 0.7.0 stack, these eight public R1-distill checkpoints are not ranked by a single pass@1 or token count. Clustered pass@1, maj@5, Both-OK versus BF16-correct conditional length, and the historical token proxy versus sequential versus batched $\widetilde{C}^{\mathrm{hyb}}_{\mathrm{pass}}$ can disagree. The tested Llama AWQ-4 checkpoint shows the largest MATH-500 and GSM8K pass@1 drops among the evaluated Llama cells; the tested Qwen AWQ-4 checkpoint shows the largest GPQA-Diamond drop among the evaluated Qwen cells (Holm-significant within the primary six GPQA contrasts; not under the exploratory 18-contrast Holm correction); FP8–BF16 MATH intervals include zero but fail ± 1 pp TOST. Qwen 4-bit MATH length rises 6.3–6.9%, with Both-OK CIs excluding zero; mismatch-conditioned differences are much larger because failure traces are themselves long. Loops are rare (25/56,408); near-cap completions are more common for Qwen 4-bit (25/24 vs. 14). Gold-free 5/5 unique-mode abstention has low observed selective risk and strictly positive Wilson upper bounds on $0/n$ cells. Cost-estimator rankings disagree across the 65 tok/s proxy, Condition A, and Condition B; Qwen FP8 Condition B is a five-rep wall-clock mixture, not a lone -36.0%. The contribution is a stack-pinned measurement of those disagreements, not a new compressor.

Artifacts

Code, patches, validation JSON, and analysis reports: Manish06N/reasoning-compression-lab. Per-cell records: `results/math500/`, `results/gsm8k/`, `results/gpqa/`. Canonical pass@1 tables: `revision_reanalysis_report.json`. Modal agreement: `modal_agreement_report.json`. Measured serving (confirmation): `results/reports/measured_serving_confirmation/`. First unconstrained timing (provenance only): `results/measured_serving/`. See `paper/ARTIFACT.md`. GPQA is a gated benchmark; public records must not republish item text.

Acknowledgment

PARAM Rudra HPC at IIT Patna (National Supercomputing Mission, C-DAC) provided compute.

A Prompt

MATH-500 and the GSM8K/GPQA QRM templates use:

```
{question}

Please reason step by step, and put your final
answer within \boxed{}.
```

with a <think> prefix on the official QRM path.

B Checkpoints

Cell	Hugging Face ID / revision
Qwen BF16	`deepseek-ai/DeepSeek-R1-Distill-Qwen-7B` `916b56a44061fd5cd7d6a8fb632557ed4f724f60`
Qwen FP8	`RedHatAI/DeepSeek-R1-Distill-Qwen-7B-FP8-dynamic` `ceb2fcd178d477cd21b92ffb43164c74165a212c`
Qwen AWQ-4	`jakiAJK/DeepSeek-R1-Distill-Qwen-7B_AWQ` `1d082b3ded4e7cefbb6b32cd5e07c7d409832816`
Qwen GPTQ-4	`RedHatAI/DeepSeek-R1-Distill-Qwen-7B-quantized.w4a16` `9559a91ed463eecda64b93a532fe5aa8cc25f2eb`
Llama BF16	`deepseek-ai/DeepSeek-R1-Distill-Llama-8B` `6a6f4aa4197940add57724a7707d069478df56b1`
Llama FP8	`RedHatAI/DeepSeek-R1-Distill-Llama-8B-FP8-dynamic` `5d548d9169c53b7bb7ef0b3bc261509d5da6e3dd`
Llama AWQ-4	`jakiAJK/DeepSeek-R1-Distill-Llama-8B_AWQ` `57360e84a0e7278c6b3abb453f2dc812b213badb`
Llama GPTQ-4	`RedHatAI/DeepSeek-R1-Distill-Llama-8B-quantized.w4a16` `faa06ed0a4a5bdab76c4df12376e0c5228c0bdfe`

AWQ-4 checkpoints are third-party community uploads; calibration corpus and quantization code revision are unknown. GPTQ-4 and FP8 are RedHatAI compressed-tensors artifacts. Dataset revision SHAs for MATH-500 / GSM8K / GPQA-Diamond were not pinned in the campaign launcher beyond the QRM task names.

Serving confirmation protocol and raw JSON live under `docs/MEASURED_SERVING_CONFIRMATION_PROTOCOL.md` and `results/measured_serving_confirmation/raw/`. Aggregates: `measured_serving_confirmation_report.json`. An earlier unconstrained timing (`docs/MEASURED_SERVING_PROTOCOL.md`, `results/measured_serving/`) is provenance only and is not mixed with confirmation means.

Table 12: Paired coverage Δ versus matched BF16 (percentage points). Intervals that exclude zero are marked *. Selective-accuracy Δ intervals are in the machine-readable modal report.

Contrast	$\geq$3/5 cov. Δ [95% CI]	$\geq$4/5 cov. Δ [95% CI]	5/5 cov. Δ [95% CI]
Qwen FP8	$+0.6$ $[-0.8, +2.0]$	-0.8 $[-2.2, +0.6]$	$+0.4$ $[-1.6, +2.4]$
Qwen AWQ-4	-0.2 $[-1.8, +1.2]$	-2.2 $[-4.0, -0.6]^*$	-2.0 $[-4.4, +0.2]$
Qwen GPTQ-4	-0.6 $[-2.0, +0.8]$	-0.6 $[-2.2, +1.0]$	-1.6 $[-4.2, +0.8]$
Llama FP8	-0.0 $[-1.8, +1.8]$	$+0.8$ $[-1.6, +3.2]$	$+2.0$ $[-1.2, +5.2]$
Llama AWQ-4	-0.8 $[-2.8, +1.2]$	-3.2 $[-6.0, -0.6]^*$	-6.0 $[-9.4, -2.6]^*$
Llama GPTQ-4	-0.2 $[-2.2, +1.8]$	-1.4 $[-3.8, +1.0]$	-0.8 $[-4.2, +2.6]$

C Reanalysis

Pathology, clustered bootstrap, and token strata are produced by the canonical reanalysis script from the released per-cell JSON. Loop threshold = 20. Near-cap threshold = 32,500. Bootstrap $B = 10{,}000$, seed 0.

D Modal-agreement details

The compact recovered artifact has 20,000 rows (SHA256 prefix `23e9ead021111959`; full digest is in the JSON sidecar). It stores extracted prediction representations, campaign `extractive_match`, and completion tokens. It does not store chain-of-thought or problem text. Campaign extraction used LightEval 0.8.0; a later MacBook LightEval 0.8.1 install was not used for these numbers. Campaign extractive match was reproduced on 20,000/20,000 rows before clustering. Equivalence clustering reported 0 symmetry violations, 0 transitivity violations, and 116 non-unique-mode ties among 4,000 groups. Bootstrap uses $B = 10{,}000$ problem resamples with seed 20260816. Secondary ECE/Brier scores (confidence = $m_i/5$) are in the JSON report and are not treated as a primary calibration result.

T_5 is the five-sample token-cost proxy: the sum of completion tokens over seeds 42–46. Mean / median / P90 T_5 are computed over all 500 items. Tokens per served query and tokens per correct served result use that same 500-item T_5 total in the numerator, so abstained queries are not free.

E Measured serving: latency, microbenchmark, FP8 repeats, and token proxy

Condition A latency detail (Table 13) is sequential decode of a balanced 20-prompt MATH-500 subset. Median latency is the mean of per-run medians. $R = 3$ except Llama AWQ-4 ($R = 5$; CV-triggered). Fixed-token microbenchmark (Table 14): 10 prompts $\times$ 512 tokens, $T = 0.0$, `ignore_eos`. Table 15 lists all five Qwen FP8 Condition B repeats. Table 16 is the historical 65 tok/s token proxy, not measured wall-clock.

F Holm-18 sensitivity

Primary multiplicity control is Holm–Bonferroni within each benchmark's six format contrasts. Table 17 applies Holm jointly to all 18 contrasts as a stricter sensitivity. Llama MATH AWQ-4 and Llama GSM8K AWQ-4 remain significant. The tested Qwen AWQ-4 checkpoint decreases GPQA-Diamond pass@1 by 5.56 pp and is Holm-significant within the primary six-contrast GPQA family; it does not remain significant under the stricter exploratory Holm correction

Table 13: Confirmation single-stream serving (Condition A: 20 sequential prompts). Latency columns are means of per-run median/P90. $\widetilde{C}^{\mathrm{hyb}}_{\mathrm{pass}}$ uses campaign MATH-500 pass@1 and a \$1.50/A100-hour scenario. Tok/s is mean± sample SD.

Model	Fmt	tok/s	Med. lat (s)	P90 lat (s)	GPU-s/q	$\widetilde{C}^{\mathrm{hyb}}_{\mathrm{pass}}$
Qwen-7B	BF16	43.92 ± 0.04	59.19	244.83	102.51	\$0.0454
Qwen-7B	FP8	62.50 ± 0.08	43.60	197.01	84.60	\$0.0373
Qwen-7B	AWQ-4	76.89 ± 0.70	26.47	110.43	48.04	\$0.0215
Qwen-7B	GPTQ-4	82.67 ± 0.69	34.78	67.51	54.39	\$0.0242
Llama-8B	BF16	72.34 ± 0.27	32.40	126.10	66.43	\$0.0310
Llama-8B	FP8	78.75 ± 2.89	38.00	253.55	92.55	\$0.0431
Llama-8B	AWQ-4	70.20 ± 2.71	40.12	246.14	89.51	\$0.0431
Llama-8B	GPTQ-4	63.49 ± 0.41	46.31	242.83	86.95	\$0.0407

Table 14: Fixed-token decode microbenchmark (10 prompts × 512 tokens). Isolates kernel decode speed from trace-length variation. Secondary to task-realistic Conditions A/B.

Model	Fmt	Raw decode tok/s
Qwen-7B	BF16	205.41
Qwen-7B	FP8	437.19
Qwen-7B	AWQ-4	377.73
Qwen-7B	GPTQ-4	401.19
Llama-8B	BF16	362.01
Llama-8B	FP8	392.05
Llama-8B	AWQ-4	344.04
Llama-8B	GPTQ-4	302.54

across all 18 benchmark contrasts (Holm-adjusted $p = 0.109$). That does not change the primary within-benchmark inference.

G Superseded analyses

The following notes are retained only so old links do not 404; they are not mixed with canonical numbers.

200-item even-index length. A previous "stratified mixed-correctness" analysis used even indices among the first 400 MATH-500 problems, seed 42 only, and the mean of per-item ratios. On those 200 items it reports Qwen FP8 +11.4% while the ratio of means is −3.4%. The canonical estimator is the full-grid ratio of means plus clustered paired Δ (Table 6).

Unconstrained serving timing. `results/measured_serving/` used unpinned `max_num_seqs`, an unbalanced Condition A first-20 draw, and mixed-node Llama repeats. Absolute tok/s from that run is not comparable to the confirmation after pinning concurrency 8. Those files are provenance; they are not averaged with confirmation and are not cited in the abstract.

Pathology key mismatch. Earlier drafts reported 0/0 loops and cap hits because analysis scripts read `truncation_count` / `repetition_flag_count` while the validator writes `token_limit_hits` / `repetition_rows`. Canonical counts are 25 loops and 0 exact cap hits over 56,408 completions (Section 4.3).

Table 15: Qwen-7B FP8 Condition B: all five technical repeats (identical aggregate 373,794 output tokens). Slow: tok/s < 400. Fast: tok/s > 500. $\widetilde{C}^{\text{hyb}}_{\text{pass}}$ uses campaign pass@1 0.944. The cause of the timing regimes was not identified.

Rep	**tok/s**	**GPU-s/q**	$\widetilde{C}^{\text{hyb}}_{\text{pass}}$	**Regime**
1	350.91	10.6522	\$0.0047	slow
2	350.68	10.6591	\$0.0047	slow
3	455.90	8.1990	\$0.0036	mid
4	545.18	6.8563	\$0.0030	fast
5	546.26	6.8427	\$0.0030	fast
Mean ± SD	449.79 ± 97.53	—	\$0.0038	keep all five
Median / IQR	455.90 / [350.91, 545.18]	—	—	—

Table 16: Historical token-implied aggregate cost proxy on MATH-500 at \$1.50/A100-hour and a shared 65 tok/s. Not measured wall-clock. Ranking equals ranking by mean tokens / pass@1. Primary cost evidence is Table 11.

Cell	**Mean tok.**	**Cost / Q**	**Token proxy**
Qwen BF16	4,011.4	\$0.02571	\$0.02736
Qwen FP8	4,007.7	\$0.02569	\$0.02721
Qwen AWQ-4	4,265.3	\$0.02734	\$0.02936
Qwen GPTQ-4	4,287.2	\$0.02748	\$0.02940
Llama BF16	4,656.6	\$0.02985	\$0.03345
Llama FP8	4,550.8	\$0.02917	\$0.03259
Llama AWQ-4	4,736.5	\$0.03036	\$0.03511
Llama GPTQ-4	4,840.5	\$0.03103	\$0.03490

vLLM 0.8.5 pathology autopsy. Earlier drafts wrote that pinning vLLM 0.7.0 "removed" truncations and loops seen on vLLM 0.8.5. Those earlier runs also differed in harness, decoding flags (including a bug that dropped `repetition_penalty`), and prompt profile. The final grid still contains 25 loops. We do not attribute pathology rates causally to the engine version. A clean stack × format factorial remains future work.

Table 17: Secondary Holm correction over all 18 pass@1 contrasts. Primary inference remains Holm-6 within each benchmark. sig.: Holm-significant at $\alpha = 0.05$; n.s.: not significant.

Bench.	**Contrast**	p	**Holm-6**	**Holm-18** p	**Holm-18**
MATH	Llama AWQ-4	< 0.001	sig.	< 0.001	sig.
MATH	Qwen AWQ-4	0.069	n.s.	1.000	n.s.
MATH	Qwen GPTQ-4	0.245	n.s.	1.000	n.s.
MATH	Qwen FP8	0.383	n.s.	1.000	n.s.
MATH	Llama GPTQ-4	0.619	n.s.	1.000	n.s.
MATH	Llama FP8	0.647	n.s.	1.000	n.s.
GSM8K	Llama AWQ-4	0.002	sig.	0.031	sig.
GSM8K	Llama GPTQ-4	0.511	n.s.	1.000	n.s.
GSM8K	Qwen AWQ-4	0.579	n.s.	1.000	n.s.
GSM8K	Qwen GPTQ-4	0.729	n.s.	1.000	n.s.
GSM8K	Llama FP8	0.761	n.s.	1.000	n.s.
GSM8K	Qwen FP8	0.827	n.s.	1.000	n.s.
GPQA	Qwen AWQ-4	0.007	sig.	0.109	n.s.
GPQA	Qwen GPTQ-4	0.288	n.s.	1.000	n.s.
GPQA	Llama FP8	0.403	n.s.	1.000	n.s.
GPQA	Llama GPTQ-4	0.573	n.s.	1.000	n.s.
GPQA	Qwen FP8	0.690	n.s.	1.000	n.s.
GPQA	Llama AWQ-4	0.711	n.s.	1.000	n.s.

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